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B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

CE 19-201-0605 CONSTRUCTION MANAGEMENT

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand principles of effective leadership.
- CO2: Read and interpret construction documents and specifications.
- CO3: Create schedules, bar charts, critical path networks.
- CO4: Identify types of float and the use of float to manage projects.
- CO5: Correlate manpower and cost loading to schedule.
- CO6: Understand the uses and working of various equipments involved in construction
- CO7: Identify all activities and issues related to planning, financing, procuring, constructing, and managing the built environment.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
✓ I. (a) What are the characteristics of an organisation?	3	L2	1	1	
✓ (b) Write notes on delegation of authority.	3	L3	1	2	
✓ (c) Write a short note on contract document.	3	L4	2	5	
✓ (d) Comment on the limitations of bar chart.	3	L2	3	1	
✓ (e) What do you mean by resource smoothing?	3	L3	4	3	
✓ (f) What are the differences between CPM and PERT?	3	L2	4	2	
✓ (g) Write short notes on pile driving equipments.	3	L2	6	7	
✓ (h) What are the factors concerning selection of an equipment?	3	L4	6	1	

PART B

(4 × 12 = 48)

✓ II.	Elaborate on characteristics of matrix organisation.	12	L5	1	3
OR					
✓ III.	Explain different types of organisation.	12	L4	1	2
✓ IV.	Elaborate the different resources of construction industry in detail.	12	L4	7	1
OR					
✓ V.	Explain the different types of contracts.	12	L4	2	4

(P.T.O.)

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Marks BL CO PO
12 L6 3 3

- VI. Determine the optimum cost and optimum duration of the project. The normal time, normal cost, crash time and crash cost are given in the table. Take indirect cost as ₹ 850/day.

Activity	Normal		Crash	
	Time (days)	Cost (in ₹)	Time (days)	Cost (in ₹)
1-2	5	600	4	900
1-3	7	1500	5	2500
2-3	3	300	2	600
3-5	6	1500	4	2500
2-4	3	1000	3	1000
1-5	9	1150	7	1750
4-5	4	1000	2	2400

OR

- VII. The activities and their expected time of a project are shown in table below. From this table draw the network and determine the activity time and floats. Also determine the critical path and its duration.

Critical Path : 1-3-4-5-6-8 Duration : 34 days

Activity	1-2	1-3	1-4	2-7	3-4	4-5	4-7	5-6	5-7	5-8	6-8	7-8
Duration (days)	9	11	6	7	4	8	0	5	4	7	6	6

- VIII. With proper sketches, explain the basic parts, operation and application of a power shovel.

OR

- IX. Explain in detail, the various factors to be considered for the selection of equipment.

Bloom's Taxonomy Level's

L2 - 11%, L3 - 6%, L4 - 32%, L5 - 30%, L6 - 21%.
